

AHTOPOL, Bulgaria

WMO: 156610

Lat: 42.100N

Lon: 27.940E

Elev: 62

StdP: 14.66

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.6	25.1	15.5	12.1	25.7	18.8	14.2	28.4	32.0	31.1	27.9	34.2	12.1	320	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.9	85.7	72.0	83.3	72.0	81.4	71.2	75.6	81.8	74.0	80.8	72.6	79.7	7.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	126.0	79.9	71.6	116.8	78.9	70.0	110.5	77.9	39.2	81.9	37.6	81.1	36.3	80.0	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.5	21.3	18.1	DB	17.0	91.2	5.1	4.2	13.4	94.2	10.4	96.7	7.6	99.0	3.9	102.1
			WB	15.7	77.5	5.0	2.4	12.1	79.2	9.2	80.6	6.4	81.9	2.7	83.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.2	39.7	40.5	46.1	51.8	60.5	68.5	73.2	73.9	67.2	58.9	50.6	42.8
	DBStd	13.71	8.29	8.62	8.12	6.34	5.75	4.69	3.34	3.68	4.25	6.23	6.91	8.45
	HDD50	1185	334	286	179	48	1	0	0	0	0	8	74	255
	HDD65	3991	785	687	587	398	162	19	0	0	27	204	434	688
	CDD50	3463	14	19	59	103	327	554	721	741	515	285	92	33
	CDD65	793	0	0	1	3	23	123	256	276	93	16	2	0
	CDH74	4857	0	0	5	16	106	672	1721	1972	293	70	2	0
	CDH80	800	0	0	0	1	17	104	284	360	18	16	0	0
Wind	WSAvg	7.5	8.1	8.3	8.0	6.8	6.2	6.1	6.9	7.2	8.0	8.4	7.3	8.6
Precipitation	PrecAvg	24.8	2.4	2.0	2.1	1.7	1.3	1.4	1.2	1.2	2.1	3.0	3.0	3.3
	PrecMax	43.6	6.7	5.0	6.0	4.4	3.3	3.4	5.3	5.2	6.7	8.5	7.4	6.7
	PrecMin	15.3	0.2	0.4	0.4	0.1	0.2	0.0	0.0	0.0	0.0	0.3	0.2	0.7
	PrecStd	6.7	1.6	1.2	1.3	1.1	0.9	1.0	1.2	1.5	2.0	2.2	1.9	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	66.1	72.7	76.0	81.7	87.8	88.4	90.5	82.7	81.0	73.3	65.9
		MCWB	53.6	53.7	56.4	58.4	67.1	70.6	72.6	72.2	68.1	64.8	59.4	56.1
	2%	DB	60.7	61.2	67.2	70.3	76.5	82.2	85.3	85.9	79.0	74.3	68.2	62.8
		MCWB	52.2	52.3	54.5	56.4	64.4	70.3	72.1	73.0	68.3	64.7	59.7	54.6
	5%	DB	55.6	57.2	62.6	65.4	72.8	79.8	83.0	83.5	77.1	70.8	63.9	58.9
		MCWB	48.7	50.1	52.3	54.7	63.4	69.2	72.5	72.6	67.8	63.1	57.5	52.6
	10%	DB	51.6	53.1	58.6	61.6	69.9	77.7	81.0	81.5	75.4	67.9	60.7	54.8
		MCWB	46.5	47.6	50.4	53.6	62.1	68.5	71.4	71.9	66.5	61.8	56.2	50.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.3	55.6	57.7	60.2	68.8	74.1	77.4	78.7	72.6	68.0	62.5	57.3
		MCDB	62.0	63.9	70.7	70.9	77.6	82.7	82.2	83.5	78.4	74.9	68.6	63.7
	2%	WB	52.2	52.5	55.3	57.3	66.5	72.1	75.6	76.2	70.8	65.5	60.3	55.4
		MCDB	59.8	60.1	65.0	67.4	74.1	79.6	81.2	82.2	76.3	72.3	66.1	61.4
	5%	WB	49.4	50.4	53.1	55.6	64.6	70.8	74.0	74.4	69.4	63.9	58.4	53.2
		MCDB	55.0	56.6	61.4	64.1	71.0	78.1	80.8	81.0	75.0	69.7	63.3	58.1
	10%	WB	46.9	47.8	50.9	54.0	62.8	69.4	72.5	73.0	67.8	62.6	56.2	50.7
		MCDB	50.8	53.1	58.1	60.6	68.7	76.5	79.8	79.9	73.6	67.5	60.7	54.8
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.5	11.4	11.7	13.1	14.0	13.6	13.9	12.6	11.0	11.1	10.1
		MCDBR	15.5	17.4	17.7	18.6	18.0	17.1	15.8	16.1	15.5	16.3	15.4	15.3
		MCWBR	10.4	10.6	9.2	8.9	10.1	9.6	9.7	9.4	9.1	9.2	9.5	10.0
	5% WB	MCDBR	14.9	16.1	16.4	16.1	15.3	15.5	14.0	14.2	13.4	13.1	13.0	13.6
		MCWBR	10.5	10.2	9.4	8.7	9.8	9.9	9.5	9.4	9.0	8.6	8.7	9.5
Clear-Sky Solar Irradiance	taub	0.365	0.397	0.432	0.477	0.444	0.443	0.448	0.457	0.440	0.413	0.386	0.371	
	taud	2.316	2.216	2.145	2.037	2.213	2.260	2.260	2.230	2.244	2.284	2.297	2.293	
	Ebn at Noon	240	250	255	252	265	265	262	256	250	242	229	225	
	Edh at Noon	28	36	43	52	44	43	42	42	39	33	28	27	
All-Sky Solar Radiation	RadAvg	458	683	1084	1535	1951	2152	2196	1941	1396	888	548	404	
	RadStd	65	84	102	155	123	147	107	91	117	89	74	62	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)		+0.97	N/A	N/A	N/A	+0.92	+1.04	-126	-202	+196	+108

Nomenclature: See separate page